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LATEST INVENTIONS IN SCIENCE AND TECHNOLOGY

HYPERSPECTRAL IMAGING:

ISRO launched its first Hyperspectral imaging satellite (HysIS) by PSLV C43 on November 29, 2018. HysIS, weighting about 380 Kg, is an earth observation satellite. The primary goal of HysIS is to study the earth's surface in the visible, near infrared and shortwave infrared regions of the electromagnetic spectrum. It will be used for a wide range of application like agriculture, forestry, soil survey, geology, coastal geology, coastal zones, inland water studies, environmental studies and detection of pollution from industries. Being an earth observation satellite, it will also be used by the military for surveillance purpose.

Hyperspectral Imaging

1. Hyperspectral imaging or hyspex imaging (imaging spectroscopy) combines the power of digital imaging and spectroscopy, collects and processes information from across the electromagnetic spectrum.
2. Hyspex' imaging enables distinct identification of objects, materials or processes on earth by reading the spectrum for each pixel of a scene from space.
3. It has the ability to add a new dimension to the usual optical imagers
4. The hyspex technology is still very rich and evolving with time, and in recent times, it has become a trend that is being experimented with globally.

Applications

The HySpex hyperspectral camera is capable of acquiring data well beyond the spectral range of the human eye, which is limited by a maximum wavelength of about 700nm. HySpex camera can be configured for imaging out to 250nm, thus including a large portion of the infrared spectrum. For many applications, the reflection/absorption properties in the IR region are essential to characterize, quantity or classify the objects in the scene.

Hyperspectral imaging is also used in various other applications – satellite-based/airborne remote sensing, military target detection, industrial quality control,

online industrial monitoring / sorting / classification, lab applications in medicine and biophysics, laboratory measurements, clinical instruments for medical diagnostic and airborne /satellite-based remote sensing tools.

LIGHT EMITTING PLANTS

A team of scientists from the University of California and the Massachusetts Institute of Technology (MT) has found a way to infuse plants with the luminescence of fireflies. By embedding specialized nanoparticles into the leaves of a watercress plant, they induced the plants to give all dim light for nearly four hours. They believe that, with further optimization, such plants will one day be bright enough to illuminate to workspace. Lighting, which accounts for about 20 percent of worldwide energy consumption, seemed like a logical next target.

'Plants can self-repair, they have their own energy, and they are already adapted to the outdoor environment'.

The light is ultimately powered by the energy metabolism of the plant itself; this technology could also be used to provide low-intensity indoor lighting or to transform trees into self-powered streetlights.

To create glowing plants, the MIT team turned to luciferase, the enzyme that gives fireflies their glow. Luciferase acts on a molecule called luciferin, causing it to emit light.

Another molecule called co-enzyme A helps the process along by removing a reaction by-product that can inhibit luciferase activity. To get the particles into plant leaves, the researchers first suspended the particles in a solution. Plants were immersed in the solution and then exposed to high pressure, allowing the particles to enter the leaves through tiny pores called stomata.

The researchers' early efforts at the start of the project yielded plants that could glow for about 45 minutes, which they have since improved to 3.5 hours. The light generated by one 10-centimetre watercress seeding is currently about one-thousandth of the amount needed to read by, but the researchers believe they can boost the light emitted, as well as the duration of light, by further optimizing the concentration and release rates of the components.

The researchers have also demonstrated that they can turn the light off by adding nanoparticles carrying a luciferase inhibitor. This could enable them to eventually create plants that shut off their light emission in response to environmental conditions such as sunlight.

Nano bionic plants

Nano bionic plants, a new research area which aims to give plants novel features by embedding them with different types of nanoparticles. The goal is to engineer plants to take over many of the functions now performed by electrical devices. The researchers have previously designed plants that can detect explosive and communicate that information to a smartphone, as well as plants that can monitor drought conditions.

SPACE-BASED INTERNET

Spaced-based internet will beam the internet data directly from the satellites in the Low Earth Orbits to the customers on the ground. It would give access to all 4 billion people who don't' have internet access yet. With enough bandwidth, such a network could become instant competitors to telecoms like AT&T and Verizon without the monumental infrastructure costs of putting down the fibre. There are eight companies currently shooting for the goal.

Some of these companies, like SpaceX, Samsung and Boeing, are well known, Others, like OneWeb, Telesat LEO, SES O3B, Iridium Next and LeoSat have less name recognition so far but will help spur competition in the field.

"4,600 such satellites operating at data rates in excess at Tb/s in LEO orbit can provide an overall capacity of one Zettabyte/month or 200GB/month" would be enough for 5 billion users worldwide.

Competition

SpaceX has named its program Starlink. Starting 2019, SpaceX wants to launch 4,425 operational satellites in LEO, far closer to the Earth than current networks. The company is working under the assumption that the closer to Earth, the fewer issues there will be with lag and latency. For comparison the United States Global

Positioning System uses 31 satellites. But SpaceX wants to send up an additional 7,500 satellites, operating even closer to Earth, to help out with speed.

One potential problem is the sheer volume of satellites. There are roughly 1,738 satellites currently orbiting the Earth, Starlink would put more than double that amount. Questions of space junk, just to name one issue, would become a serious issue.

Space X has one obvious advantage: It has its own rockets from which to launch the satellite. It has launched the first two Starlink test satellites (named Tintin A and Tintin B).

One Web's had a deal with Jeff B Bezos' Blue Origin for launching satellites. For One Web the general plan is to get 720 small satellites into the LEO (Low Earth Orbit) with market service starting over Alaska in 2019. Eventually, it would get 900 second-generation, higher-orbiting satellites by the mid-2020s to increase speeds. Boeing is planning with Apple to create a satellite network that would be a tremendous boost to iPhones everywhere.

Kessler Syndrome

There's a theory among astronomers called the Kessler Syndrome, named for the NASA astrophysicist who proposed it in 1978. In this scenario, an orbiting satellite or some other piece of material accidentally strikes another and breaks into pieces. These pieces whirl around the Earth at tens of thousands of miles per hour, destroying everything in their path, including other satellites. It starts a catastrophic chain reaction that ends in a cloud of millions of pieces of non-functional space debris that orbits the planet indefinitely.

Such an event could make an orbital plane functionally useless, destroying any new satellites sent into it and possibly preventing access to other orbits and even all of space.

Other related technologies

Space X's Starlink proposal is not the first-time people have tried to build an internet in the sky. Some of the other related technologies are.

1. **Project Loon**
2. Owned by Google's parent company, Alphabet, Project Loon is building a network of high-altitude balloons to beam the internet down to remote regions from the stratosphere. Earlier this year, Loon graduated from a 'moon-shot project" to become its own company. It is now working with a telecoms firm in Kenya.
3. Facebook has been testing a fleet of solar-powered drones with wingspans larger than a Boeing 737 that shoot the internet at the ground with lasers. The idea is to try to get underdeveloped countries online. After several setbacks, including at least one drone crash, Facebook abandoned the project.
4. Facebook is developing a satellite house named Athena specifically to offer internet services to underdeveloped areas.
5. A handful of other companies, including O3b Networks and Iridium have been serving the internet to remote places from space for more than a decade. But each offers only limited coverage via a few dozen satellites.

Implementation

1. Satellites in low-earth trajectories operate 50 to 1,200 miles (80 to 2,000 kilometres) above Earth and orbit it roughly every 90 minutes. Traditional communications satellites operate much higher, at an attitude of about 22,000 miles (36,000 km) and appear to hover in one spot because their orbit takes one day, matching the earth's rotation.
2. Lower satellites have an advantage in sending and receiving broadband since signals suffer less of the lag time that can interrupt phone conversations and streaming video. A universal broadband has become a more compelling goal, the appetite for service by satellites has grown.
3. When sending an internet message via Starlink, a ground station will begin by using radio waves to talk to a satellite above it. Once in space, the message will be fired from satellite to satellite using tasers until it is above its destination. From there, it will be beamed down to the right ground station using radio waves again. Between distant places, this will allow messages to be sent about twice as fast as through the optical fibres on Earth that currently connect the internet,

despite having to travel to space and back. This is because the speed of the signal in the glass is slower than it is through space.

4. On the ground, customers will mount a new sort of terminal with electronically steered antennas that automatically connect to whichever satellite is currently offering the best signal-similar to the way a cell phone picks towers. Because LEO satellites move relative to the Earth, the system will switch between them every 10 minutes or so. And because thousands will be up there, at least 20 will always be available to choose from.

If the goal is to provide (good) internet access where previously there was none, satellites are a reasonable way to achieve it. In fact, companies have been doing this for decades via large geostationary (GSO) satellites that sit in a very high orbit, fixed above a certain point on the Earth. But aside from a few niche applications, including cargo tracking and providing internet to military bases, this kind of satellite connectivity has not been fast, reliable, or responsive enough to be competitive with modern fibre or cable-based internet.

Lithium-Ion Battery

Lithium-ion batteries (Li-Ion) are a family of rechargeable batteries having high energy density and commonly used in consumer electronics. Unlike the disposable lithium primary battery, a Li-Ion uses intercalated lithium compound instead of metallic lithium as its electrode.

Usually, Li-ion is significantly lighter than other kind of rechargeable batteries of similar size. Li-Ion is heavily used in portable electronics. These batteries can be commonly found in PDAs, iPods, cell phones, laptops, etc.

When a Li-Ion battery is discharging, lithium ions move from the negative electrode (anode) to the positive electrode (cathode). When a Li-Ion is charging, lithium ions move in the opposite direction, and the negative electrode becomes the cathode, while the positive electrode becomes the anode.

Some of the advantages of a Li-Ion battery are:

1. A typical Li-Ion battery can store 150 watt-hours electricity per kg of battery, compared to 100 watt-hours electricity in a nickel-metal hydride (NiMH) battery, and only 25 wall-hours electricity in a lead-acid battery.
2. Li-Ion battery holds a charge well. They usually lose approximately 5% of their charge each month, against a 20% monthly loss for NiMH batteries.
3. Li-Ion battery does not require complete discharge before recharging
4. Li-Ion battery can handle more charge/discharge cycles.

Some of the disadvantages of Li-Ion Battery are:

1. Li-Ion battery starts to degrade the moment they leave the factory. They usually last for only two to three years from the date of manufacture, regardless of whether used or unused.
2. Li-Ion battery is highly sensitive to higher temperatures. Higher temperature leads to a much faster degradation rate than normal.
3. If a Li-Ion battery is fully discharged, it gets totally damaged.
4. Li-Ion battery is comparatively expensive.
5. There exists a small possibility that if the Li-Ion battery pack fails, it may burst open into flame.

The advantages offered by the Li-Battery technology and the technology is in its infant stage, and there are improvements that are expected. All these factors make lithium-ion batteries the future of electric vehicle batteries.

Significance of Li-Ion Batteries for India

NITI Ayog in 2017 announced that Government of India is planning to replace all the cars and buses with electric vehicles by 2032 in a bid to reduce both pollutions, to fulfill INDC targets and to reduce country's dependence on oil.

Challenges

India needs five times the current world production of Lithium to achieve its target of going for all-electric cars by 2030. But India doesn't have Lithium ore reserve which can be mined for extraction of Lithium.

Globally countries like Chile, China, Australia and Argentina have the majority of global lithium reserves.

1. Chile 7500000 MT
2. China 3200000 MT
3. Australia 27000000 MT
4. Argentina 2000000 MT
5. Portugal 60000 MT

Other countries having some lithium reserve are Brazil, U.S. and Zimbabwe etc.

China has collaborated with Australia and Chile for the development of Lithium production and effectively controls more than half the global reserves. India also needs to have such collaborations in order to procure the lithium for its highly ambitions targets.

Other Challenges are

1. **Infrastructure** – we need to develop charging infrastructure for smooth operations of the electric vehicles.
2. **Is it really clean** – The electric energy in India is majorly produced by Thermal power plants.
3. **Alternate technology** – We need to give considerable attention to alternative technologies like hydrogen-based vehicles, instead of rushing to all-electric vehicles.

CLOUD SEEDING:

A rainstorm occurs after moisture collects around naturally occurring particles in the air like dust sand, causing the air to reach a level of saturation point at which it can no longer hold in that moisture and droplets fall in the form of rain. Cloud seeding essentially helps that process along, providing additional “nuclei” around which water droplet can reside, and condensation occurs. These nuclei can be salts, dry ice or silver iodide. The most common chemicals used for cloud seeding include silver iodide, potassium iodide and dry ice (solid carbon dioxide). Liquid propane, which expands into a gas, has also been used. This can produce ice crystals at higher temperatures than silver iodide. Alter promising research, the use of hygroscopic materials, such as

table salt, is becoming more popular. During cloud seeding, increased snowfall takes place when temperatures within the clouds are between 19 and 4 ^{0}F (-7 and -20^{0} C). Introduction of a substance such as silver iodide, which has a crystalline structure similar to that of ice, will induce freezing nucleation. There are three methods by which cloud seeding is done:

1. Static cloud seeding
2. Dynamic Cloud seeding
3. Hygroscopic cloud seeding

Advantages of Cloud Seeding:

1. Creates Rain: When rain is badly needed, cloud seeding is perhaps the only way to produce rain. People use silver iodine to induce rain production in areas where there a barely any precipitation. Rain is important for keeping the area hydrated and fertile for growing crops and other plants.
2. Boosts Economy: Where there is rain, there is farm produce. Farms that yield better can help the local economy and feed the people (and even animals). Cloud seeding can greatly improve the living conditions in dry, arid places.
3. Regulates Weather: Cloud seeding in a way gives us the ability to control the weather condition in a particular area. It does not just make rain; it also regulates water vapour that in turn prevents damages brought by destructive hails and storms.
4. Makes Dry Places More Livable: Local people have an impressive way of adapting to their natural environment. But inhospitable places rarely visited by rain can be inhospitable to tourists and foreigners. Cloud seeding can make such places livable.

Disadvantages:

1. Requires Potentially Harmful Chemicals: Chemicals used in cloud seeding can potentially damage the environment, especially the plant's cloud seeding is intended to protect. There is no substantial study done on the implications of silver iodine on the environment. Silver iodine may cause – iodism, a type of iodine poisoning where the patient exhibits running nose, headache, skin rash, anaemia, and diarrhoea, among others. It is highly toxic to fish, livestock and humans.

2. Is Not Fool Proof: Cloud seeding requires rainclouds. It cannot work on just any other cloud formations. Also, seeded clouds may travel to another location and do not cause precipitation on he intended location. Therefore, it can be argued whether or not cloud is truly effective in producing rain.
3. Costs a Lot: It is very expensive to produce artificial rain. The chemicals have to be delivered to the air via planes, which are hard to come by in places with very minimal income, Poverty-stricken areas suffering drought or famine may need external funding to have cloud seeding.
4. Poses Weather Problems: If not regulated or controlled properly, cloud seeding may cause undesirable if not altogether destructive weather conditions such as flooding, storms, hail risks, etc. Places that naturally do not get much rain or no rain at all usually do not have the infrastructure to handle so much precipitation. With cloud seeding, these areas may become flooded quickly, causing more harm than good.

More the water resource of a nation the higher will be the opportunities for achieving high rates of progress in the agriculture sector and industrial growth. Indian water resources are declining, and India is going to be water stressed country till 2025. 16 percent of the total area is drought prone and farmers are committing suicide, so cloud seeding becomes the necessity to tackle this critical situation. The cloud seeding technology is very closely associated with water resource management – a major requirement for India.

NANO BUBBLES

Bubbles are gas-filled cavities within liquids (and solids; here we deal with liquids only). In liquids bubbles have internal equilibrium pressures at least that of the external environment. Each bubble is surrounded by an interface that possesses different properties to that of the bulk solution.

Nanobubbles are generally recognized in the current scientific literature as those gaseous cavities with diameters less than a micron. As such cavities (bubbles) are often greater than 100 nm in diameter but the term ‘nano’ is applied mostly to particles of smaller diameter.

Nanobubbles (ultrafine bubbles) are sub-micron gas-containing cavities in aqueous solution.

The growing significance of nanotechnology has now led to the development of nanobubbles, which a drawing a lot of attention from a variety of fields. Nanobubbles play a vital role in many sectors that include scientific research, medical and food sectors.

Properties of Nanobubbles

The small size of nanobubbles gives them much more interesting properties than larger bubbles, due to their high specific area and high stagnation features in the liquid phase. Free-radical generation occurs when micro and nano-sized bubbles collapse due to the high density of ions at the gas-liquid interface that develops prior to the collapse.

It has been shown that nano-bubbles cannot remain in a stable state under atmospheric pressure due to extremely high internal pressure. Following a number of experiments, it has been suggested that the typical lifespan for a gas bubble with 100 nm radius would be around 10 μs.

In saturated liquids, these nano-bubbles are extremely stable to the absorption of ions on their surface. The gas molecules inside the nanobubbles do not come in contact with the bulk liquid. Allowing the nanobubbles to last for a much longer time. Larger bubbles have air that is above atmospheric pressure, but nanobubbles contain internal pressures of tens or even hundreds of atmospheres. Under such high pressures, it has been proven that the liquid absorbs the gas inside the nanobubbles.

Application of Nanobubbles

1. Acceleration of metabolism in vegetables and shellfishes
2. Microfluidics
3. Water treatment by flotation -their high specific area makes them useful in this field
4. Contrast agents for ultrasonography
5. Sterilization using ozone gas

6. Foam products in the food industry and other products requiring bubble stability
7. Nutritional supplement carrier in the food industry.

Recent Developments

Recently, the medical field has also seen major discoveries relating to nanobubbles. Researchers from the University of Texas MD Anderson Cancer Centre, Rice University and Baylor College of Medicine the concentrating on the development of new techniques to directly inject genetic payloads and drugs into cancer cells.

NANOZYME

A nanozyme is any inorganic nanoparticle (bare or surface modified) that mimic the catalytic properties of various enzymes and thus the name 'Artificial Enzyme'. A typical example is a Fe304 nanoparticle which is well known for mimicking the peroxidase enzyme. This means this nanoparticle can catalyze the same reactions originally catalyzed by peroxidase enzyme in biological systems. Nanozyme possess various advantages over natural enzymes such as high stability, case of storage, easy synthesis etc. However, turn over number and selectivity are not high enough to compare with natural enzymes. There are a large number of nanozymes that are being discovered. The list is long, some of them are listed below.

1. Gold nanoparticles, Peroxidase and Glucose oxidase properties
2. CeO_2 nanoparticles, Peroxidase
3. Platinum nanoparticles, Superoxide dismutase
4. CdS-Pt nanoparticle Nitrate reductase.

Applications:

1. Nanozyme to catalytically reduce hydrogen peroxide, as high levels have been reported to generate reactive oxygen species and subsequently induce oxidative stress, which can damage the DNA, proteins and lipids. The enzyme glutathione peroxidase maintains the levels of hydrogen peroxide in our body and prevents cell damage. But under oxidative stress condition, the amount of enzyme is not-sufficient to maintain the hydrogen peroxide level. Under these circumstances, the nanozyme that precisely functions as the natural enzyme can be used.

2. The nanozyme uses the same pathway as the natural enzyme but without generating any free radicals.
3. Nanozymes with tunable catalytic properties is emerging as the next generation of artificial enzymes that find application in neuroprotection, cardio protection and cancer therapy.

Nanomaterials that can behave like human enzymes have now been successfully synthesized by a team of researchers from the Indian Institute of Sciences (IISc), Bengaluru. They produced the new nanozyme – nanomaterial with enzyme-like activity by using vanadium pentoxide nanocrystals of just 150-200 nm size.

The nanozyme was able to act as the natural antioxidant enzymes glutathione peroxidase in our body and help maintain the hydrogen peroxide levels within the threshold. They synthesized the nanozyme with four different morphologies – nanowires, nanosheets, nanoflowers and nanospheres.

Researchers in China have developed a new nanozyme test for faster and cheaper testing of the Ebola virus. This test uses magnetic nanoparticles, which works like enzymes to make the signal stronger, giving a clearer result one can see with the naked eye.

e-DNA:

Environmental DNA (eDNA) – is defined as genetic material obtained directly from environmental samples (soil, sediment, water, etc.) without any obvious signs of biological source material – is an efficient, non-invasive and easy-to-standardize sampling approach. Coupled with sensitive, cost-efficient and ever-advancing DNA sequencing technology, it may be an appropriate candidate for the challenge of biodiversity monitoring.

Extraction Process

Sources of eDNA include secreted faeces, mucous, gametes, shed skin, hair and carcasses. Recent research has shown that the DNA of a range of aquatic organisms can be detected in water samples at very low concentrations using qPCR **(quantitative Polymerase Chain Reaction)** methods. Environmental DNA has been obtained from ancient as well as modern samples and encompasses single species detection to

analyses of ecosystems. eDNA has a lot of applications. E.g. eDNA extraction has confirmed the presence of rare freshwater species in temple ponds in Assam. It is also used to genetically detect and identify other extant species that could influence conservation efforts.

E-DNA method allows for bio-monitoring without requiring collection of the living organism, creating the ability to study organisms that are invasive elusive, or endangered without introducing anthropogenic stress on the organism. Access to this genetic information makes a critical contribution to the understanding of population size, species distribution, and population dynamics for species not well documented. The integrity of eNDA samples is dependent upon its preservation within the environment. Soil, permafrost, freshwater and seawater are well-studied macro environments from which eDNA samples have been extracted, each of which includes many more conditioned sub-environments. Because of its versatility, eDNA is applied in many sub-environments such as freshwater sampling, seawater sampling, terrestrial-soil sampling (tundra permafrost), aquatic soil sampling (river, lake, pond and ocean sediment), or other environments where normal sampling procedure can become problematic.

3D PRINT DRUG DELIVER

The US Food and Drug Administration approval of the first 3D printed tablet in 2015 has ignited growing interest in 3D printing, or additive manufacturing (AM), for drug delivery and testing systems. Beyond just a novel method for rapid prototyping. AM provides key advantages over traditional manufacturing of drug delivery and testing systems. These include the ability to fabricate complex geometries to achieve variable drug release kinetics; ease of personalizing pharmacotherapy for the patient and lowering the cost for fabricating personalized dosages. Furthermore, AM allows the fabrication of complex and micron-sized tissue scaffolds and models for drug testing systems that closely resemble in vivo conditions. However, there are several limitations such as regulatory concerns that may impede the progression to the market. Here, we provide an overview of the advantages of AM drug delivery and testing, as compared to traditional manufacturing techniques. Also, we discuss the key challenges, and future directions for AM enabled pharmaceutical applications.

It has wide applications. A pharmacist could print and dispense personalized medication through an automated 3D printing system, and the dosage can be adjusted based on clinical response.

The genesis of 3D printing technology can be traced back to 1986, when an American engineer, Chuck Hull, created a prototype for a process called stereolithography. Often referred to as 'the father' of 3 D printing, Hull laid down the foundation on which the modern 3D printing technology is built upon. From increasing cost-effectiveness and efficiency to spurring innovation, technology has made a significant impact on the manufacturing industry in the past couple of years. Here are live revolutionary applications of 3D printing technology and how they are transforming our daily lives.

Printing a Human Hearts

3D-printing human organs for transplants may soon become a reality. BIOLIFE4D, a biotech startup, is soon going to open a lab in Chicago and begin perfecting the process of 3D-printing human hearts, that could eventually be used in transplants. Working with researchers in university labs, the technology of use the patient's own cells, in what is called bio-ink, to produce a patient-specific, fully-functioning heart.

Questions:

1. Discuss in detail about Lithium-ion battery with its significance.
2. Mention some of the recent inventions in the field of science and technology. Explain it
3. Give a note on
   a) Nano bubbles b) Cloud Seeding.